

Robust Charging Infrastructure Design under Demand Uncertainty

A solar-powered charging station for a shared e-micromobility fleet — Executive Summary

Competition Themes — 3 (primary): solar-PV charging-station design · 2 (secondary): charge/discharge profiles under demand scenarios

25 kWp PV · 50 kWh · 4 chargers Recommended robust design	-60% (£31,574/yr) Value of robustness (worst-case cost)	99% (vs 87% naive) Guaranteed fleet service, worst case	84% (6.2 tCO2/yr) Operational carbon saving
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1. Problem Understanding [20% criterion]

Shared e-micromobility fleets face highly variable charging demand — driven by season, weather, local events and multi-year growth. This creates a sizing dilemma for a solar charging station: a station built for **average** demand fails at peaks and strands the fleet, while one built for the **worst case** sinks capital into rarely-used equipment. The dilemma sharpens under a **constrained grid connection** — upgrading the connection needs a costly network reinforcement — so the evening charging peak must be served from on-site solar and storage. Sizing therefore becomes a decision under deep uncertainty: under-build and lose service; over-build and waste money. The demand range is grounded in the real national e-scooter trial monitoring dataset (monthly trips, fleet size, daily utilisation, trip distance), so the problem is posed on observed data rather than assumption.

2. Relevance & Innovativeness of the Proposed Solution [20% criterion]

The solution directly delivers **Theme 3** (design of a solar-PV charging station for a shared fleet) and **Theme 2** (modelling charge/discharge profiles under demand scenarios), producing an actionable station specification with uncertainty bounds. Its novelty is methodological: charging-station sizing is almost always **deterministic** (a single demand forecast). We instead apply **robust optimisation — minimax-regret and maximin — alongside a chance-constrained stochastic program**, choosing a design that performs across the entire demand space, and we define and quantify the “**value of robustness**”: the worst-case cost and lost service avoided by hedging. A correlated Monte-Carlo “demand regime” reproduces the realistic co-movement of trips, utilisation and growth, and exposes a subtle but important result — expected-value design **under-weights the correlated high-demand tail**, which is precisely why scenario-based robust methods are required here.

3. Approach — Methodology, Algorithm & Flow [20% criterion]

The pipeline (Figure 1) is: **(i)** build Low/Medium/High demand scenarios from the real trial data, scaled across a year-on-year growth range into nine probability-weighted scenarios; **(ii)** simulate an 8,760-hour energy balance for any design under the capped grid, dispatching PV → battery (with off-peak grid pre-charging for peak-shaving) → capped grid → unmet demand; **(iii)** evaluate all 150 designs (PV 5–25 kWp × battery 0–50 kWh × 4–20 chargers) against the nine scenarios, pricing unmet demand as lost service, to form a cost matrix; **(iv)** apply four decision rules — naive deterministic, chance-constrained stochastic, minimax-regret and maximin — and map the cost-vs-robustness Pareto frontier; **(v)** propagate uncertainty through a 500-sample correlated Monte-Carlo fan; **(vi)** rank the drivers with both a tornado (one-at-a-time) and a variance-based global sensitivity analysis (total-effect Sobol indices, Saltelli estimator). The hourly controller is a transparent greedy priority rule (optimal for a single store) and the full study runs in about 30 seconds.

Methodology — Robust Solar Charging-Station Design

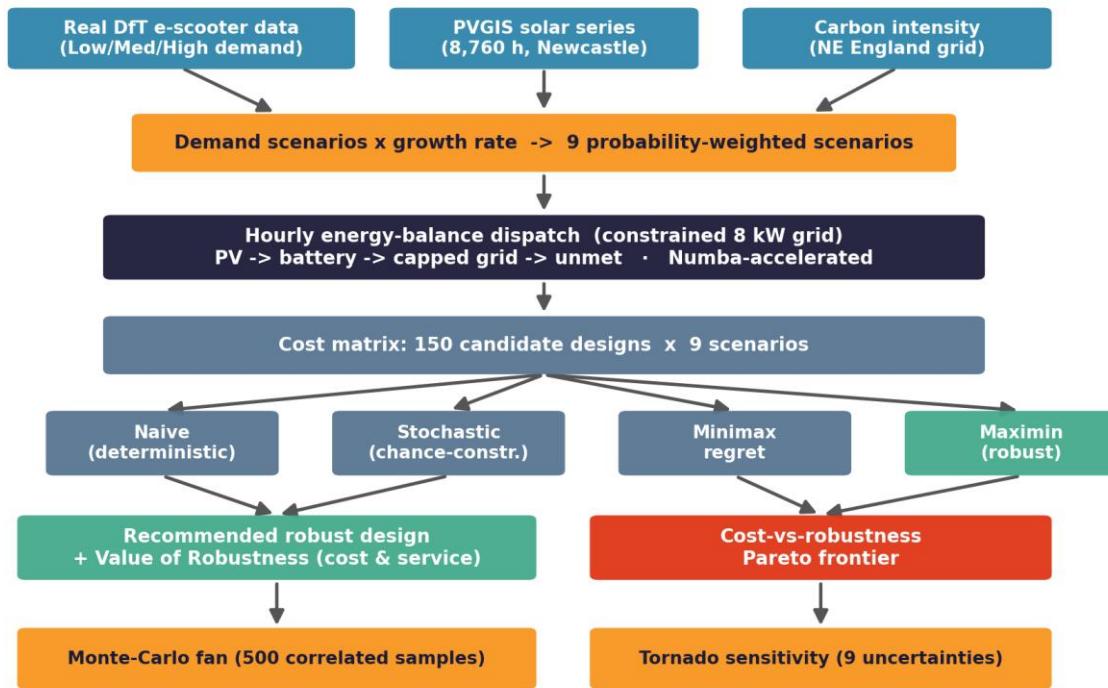


Figure 1 — End-to-end methodology / algorithm flow.

4. Feasibility of the Proposed Approach [20% criterion]

Technical: the design uses only commercially available components — monocrystalline PV, a Li-ion battery (0–50 kWh) and standard AC charge points — sized within ranges supported by manufacturer datasheets and industry benchmarks. **Economic:** the recommended robust station has a ≈£68,400 capital cost and a bounded annual cost even in the worst demand scenario; the model reports CAPEX, OPEX, battery-replacement and grid costs. **Operational:** it yields one deployable specification plus the confidence interval around it. **Computational:** the entire model is open-source Python, deterministic, regenerates all data and figures from a single command, and is covered by an automated regression-test suite — so any reviewer can reproduce and audit every number.

5. Originality & Authenticity (AI / Plagiarism) [20% criterion]

All model code, the dispatch engine, the optimisation formulation and every figure were written from scratch for this submission. The only redistributed dataset is the national DfT e-scooter monitoring data, reused under the Open Government Licence; the solar and grid-carbon series are physically modelled and calibrated to published references (with an optional live-API path), not copied. Every numerical assumption carries an inline source citation and all literature is referenced. The repository is fully self-contained and reproducible — fixed random seed, one-command rebuild, passing tests — directly supporting the competition’s AI and plagiarism checks. No third-party text or code is reproduced.



Figure 2 — Cost-vs-robustness Pareto frontier (four decision rules).

Figure 3 — Expected vs worst-case annual cost by rule.